New York University
Computer Science Department
Introduction to Software Development
Dr. Nader Nassar

Homework#2: Expressions, assignment statements and arithmetic operators in Java

Deadline: *See NYUclasses for the deadline, 15% off per day after the deadline (3 days maximum).*

Learning Objectives:

- Writing simple programs in Java
- Learning to explain basic programming concepts
- Understanding Java expressions and assignment statements
- Learning how to work with variables in Java
- Learning the process of going from simple algorithms to implementation

Please read the guidelines carefully to avoid receiving a zero on the HW:

- Please use the following naming convention "ExerciseX.java", where X is the number of the exercise.
- Apply this convention to all future homework assignments during the semester.
- Compile and run the program for each exercises.
- Make an archive (zip file or compressed file) with all the java files (the .java files NOT the .class files) and post it on NYU Classes (StudentName_Homework_X.zip)
- It is your responsibility to make sure that the Zip files have your actual files. You may send the file to yourself by email to double check before you upload on NYU classes.
- If the graders cannot open the file, you will receive a grade of zero.
- If you send the .class files instead of the .java files (source files) you will
- receive a zero.
- Cheating will be severely addressed with an immediate zero on the homework and a report to the academic advisor and the administration.
- You will automatically lose 50% of the points for an exercise if the program does not compile and run correctly

Exercise 1 (40 pts)

Write a java program that displays the result of the following formula: where X, Y are non-zero values entered by users. *(Hint: use scanner)*

$$\frac{3+4x}{5}-\frac{10(y-5)}{x}+9\left(\frac{4}{x}+\frac{9+x}{y}\right)$$

Exercise 2. (10pts)

Write a Java program to declare two integer variables, one float variable, and one string variable and assign 20, 3, 14.6, and "Hello 101 " to them respectively. Then display their values on the screen**.**

Exercise 3 (20pts):

If $1 Dollar equals to 0.9 Euros
Write a java program that ask the user to enter the amount they need to convert.
Then the program will ask the user to enter either 1 or 2
If the user enters 1, then you convert the values from dollars to euros
If the user enters 2, then you convert the values from euros to dollars.
The output of the program is a single line
You entered xx.xx in "Dollars/Euros" based on the user and they are equals to zz.zz (Euros/Dollars)

Exercise 4 (20pts)

Write a program that predicts the score needed on a final exam to achieve a desired grade in a course. The program should interact with the user as follows:

Enter desired grade> B
Enter minimum average required> 79.5
Enter current average in course> 74.6
Enter how much the final count as a percentage of the course grade> 25
You need a score of 94.2 on the final to get **B**

Text

Relevant Formula
Score needed = minimum average required - (current average * (1- final percentage decimal)) / final percentage decimal

Exercise 5 (10pts):

Write a program that calculates the addition, subtraction, division, multiplication, and the remainder of two numbers (Integers) being entered by the user.

Data Requirements

Problem input

X /* first number */

Y /*second number */

Problem output

Sum of x and y

Subtraction of x and y

Division of x and y

Multiplication of x and y

Reminder of x and y